ANUJ RAWAT

Pune, Maharashtra

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Career Objective

Electronics Engineering student with strong skills in circuit design, embedded systems, and efficient problem-solving. Detail-oriented and effective in team environments and good in communication skills. Seeking to apply hands-on experience in microcontrollers and Python implementation to build efficient hardware and software workflows.

Experience

➢ **Hardware & Embedded Systems Intern -Cruxx Solutions *(April 2026 – July 2026)***

- Designed a 410W multi-rail BEC (19V/12V/5V) for an industrial drone using LM5116/TPS54331 ICs, and prototyped a custom 4-in-1 ESC with 3-phase MOSFET commutation and AM32 firmware on STM32/AT32 MCUs
- Built a UAV SDR datalink on LimeSDR — a 1.3 GHz TDMA/π/4-DQPSK architecture with GNU Radio flowgraphs for drone and ground station, including live video streaming and RF bench testing
- Authored technical documentation (architecture briefs, BOMs, link budgets, security architecture) and an IEEE-structured SRS for a YOLOv8n + ByteTrack object detection pipeline targeting NVIDIA Jetson Orin NX

Projects

➢ **Gamified Learning Platform Web Development, UI/UX, Game Logic, Smart India Hackathon**

- Developed a gamified learning platform website for the Smart India Hackathon (SIH) to enhance educational engagement.
- Successfully reached the 2nd round of the national-level competition with a certified participation record.
- Focused on creating interactive modules to simplify complex learning concepts through game-based mechanics.

➢ **Arduino, Soil Moisture Sensors, Submersible Pump, Motor Control**

- Developed an automated irrigation system using an Arduino microcontroller to monitor soil health.
- Integrated soil moisture sensors to accurately detect over-irrigated or under-irrigated conditions.
- Programmed a logic-based system to trigger a motor and pump to deliver water precisely when the soil required hydration.

➢ **Rain Detection System Python, Raspberry Pi 5, Machine Learning, Image Classification**

- Developed a rain detection system using a Raspberry Pi 5 and Raspberry Camera.
- Implemented a Machine Learning model for image classification to identify weather conditions using python.
- Connected the system to a DIY server to automate text alerts regarding weather factors.

➢ **Automated Home Setup IoT, ESPNODE (ESP32), Smartphone Integration, Hardware Interfacing**

- Designed an automated Internet of Things (IoT) setup with a wide range of operational capabilities.
- Utilized ESPNODE microcontrollers and various hardware components to create a smart home model.
- Developed functionality to control lighting and fans remotely via a smartphone.

➢ **Smart Street Light System — C8051F340**

- Designed an automated smart street lighting system using the C8051F340 microcontroller, an LDR, an IR sensor, and 3 LEDs.
- Implemented logic to detect nighttime conditions (via LDR) and object/vehicle presence (via IR sensor), dynamically varying LED brightness in response.
- Programmed embedded sensor interfacing and PWM-based brightness control in Embedded C.

➢ **HI 21cm Galactic Radio Observation & Galactic Mapping — Pune Single-Dish Radio Telescope**

- Drift-scan observations of two galactic fields (Scutum Arm l≈30°, Galactic Centre / Sgr A*) — 3.82 hours, 260 spectra
- Detected three spiral arm components: Local (+18.2 km/s), Sagittarius (+36.8 km/s), Norma (+111.9 km/s); peak SNR 5.1

via ν160 scan stacking
- Peak SNR 5.5 toward Galactic Centre, integrated HI flux 0.777 K·km/s; 2nd-order polynomial baseline subtraction applied
- Generated l-v diagrams, average spectra plots, and sky coverage maps; structure consistent with Burton & Gordon (1978) templates.

➢ **ESA PROBA-2 Satellite Reception & Scientific Data Analysis**
- Received PROBA-2 telemetry via Cosmos Club ground station; decoded beacon signal and analysed Doppler shift for orbital velocity and pass geometry
- Characterised pass elevation vs SNR profile; validated reception windows against TLE-based predictions
- Analysed SWAP EUV solar imagery (17.4 nm) and LYRA UV/X-ray irradiance data; studied solar coronal structures across observation windows
- Cross-referenced SWAP imagery with LYRA curves to identify solar activity signatures; documented as part of Cosmos Club's solar programme.

➢ **Real-Time ADS-B Aircraft Tracking & ATC Display**
- Decoded Mode-S and ADS-B signals using dump1090 with RTL-SDR, tuned 1090 MHz antenna, and LNA for maximum range
- Deployed tar1090 web-based ATC map on Raspberry Pi — real-time altitude, velocity, heading, and squawk code display
- Fed live decoded data to FlightRadar24's global network; recognised as an official FR24 data contributor with active station in Pune
- Remote monitoring via VNC; system tracked 50+ aircraft simultaneously

➢ **Automated Geostationary Satellite Reception & Image Archive — Raspberry Pi 5**
- Receives and decodes weather imagery from NOAA-15/18/19, Meteor-M2-3, Fengyun-2H, and GK-2A; demodulates APT, LRPT, and L-band signals via SatDump
- Deployed 24/7 headless pipeline on Raspberry Pi 5 — SatDump runs as persistent background service, auto-decoding Lband downlink from Fengyun-2H and GK-2A with no operator input
- Bash/Python automation script organises decoded imagery into date/time-stamped folder hierarchies for systematic archiving
- Fixed-dish setup exploits geostationary geometry — no rotator or TLE tracking required for continuous round-the-clock reception

➢ **APRS-Based Telemetry Transmission System**
- Built an APRS network for real-time sensor telemetry over VHF/UHF using DireWolf and packet radio protocols
- Enabled remote environmental data monitoring; designed for low-power, long-range deployment in remote locations

➢ **Mobile Ground Station — Rapid Deployment SDR**
- Portable SDR station with LNA, bandpass filters, turnstile and dish antennas; integrated HamClock on Raspberry Pi for satellite pass scheduling via VNC

Education

MIT-World Peace University — **2024 – Present**
Bachelor of Electronics and Communication Engineering — *Pune, Maharashtra*

Tagore International School — **2021 – 2023**
Higher Secondary Education – Physics, Chemistry and Maths — *Mansarovar, Jaipur*

Tagore International School — **2021**
Secondary Education — *Mansarovar, Jaipur*

Technical Skills

➢ **Languages:** C (Intermediate), Embedded C, Python, MATLAB, R (Beginner)

- **Developer Tools & Frameworks:** GNU Radio, Multisim, Keil µVision, Arduino IDE, VS Code, Linux, Raspberry Pi OS
- **Hardware:** Microcontrollers (STM32, ESP32, Arduino, Raspberry Pi), Circuit Design & Simulation (LTspice, Multisim), PCB/Power Electronics (BEC/ESC design)
- **Productivity:** Google Workspace (Docs, Sheets), Microsoft Office

Achievements and Extracurriculars

- **President, Cosmos Astronomy Club:** Leading the college's astronomy club; previously a core team member; organized Celestia '26.
  - Formerly A Project Manager now promoted to President for the academic year 2026-27.
- **HackMITWPU '25 (Cosmotron):** Led a team to 5th place; also served as an event organizer
- **Smart India Hackathon:** Round 2 Qualifier for a gamified education solution
- **Astrophotography & Observatory Operations:** Conducted deep-sky imaging sessions using automated mount/imaging software (NINA, KStars/Ekos) and telescope equipment; worked in a high-end ground station/observatory